

Additional file 1

PoolParty: streamlined design of DNA sequence libraries in Python

Supplementary methods, figures, and tables

Supplementary Methods

Benchmarks were run under Ubuntu 22.04.3 LTS in Windows Subsystem for Linux 2 on an Intel Core Ultra 9 185H workstation with 31 GB RAM allocated to WSL2, using CPython 3.12.2, PoolParty 0.1.0, and StateTracker 0.1.0. Timings are single-threaded wall-clock measurements of `generate_library`, reported as mean $\pm$ SD over five replicate runs. Peak memory is the maximum resident set size reported by `getrusage`, with each data point measured in a fresh process.

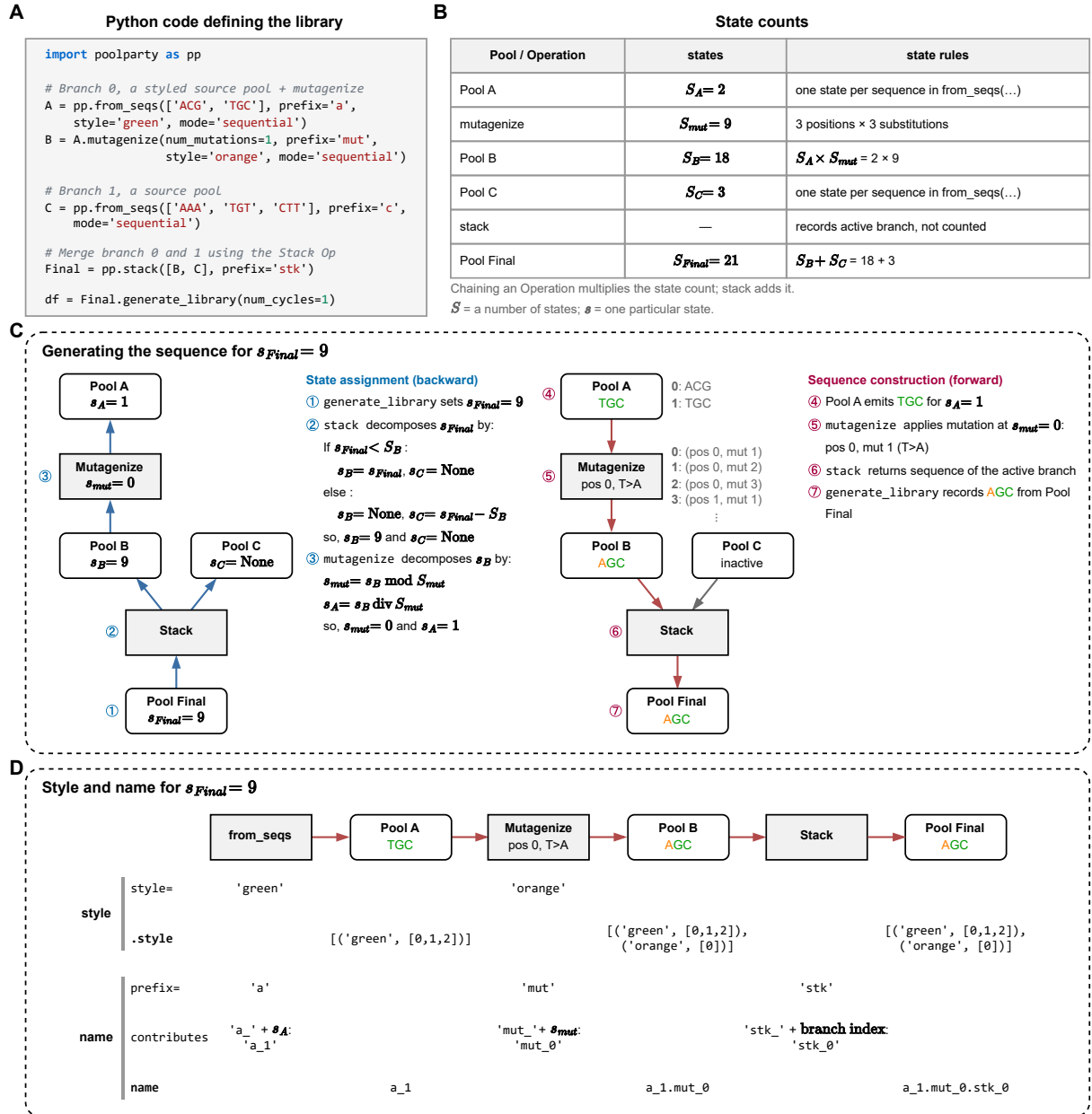


Figure S1: **A worked example of sequence generation.** (A) Python code defining the library. (B) Number of states of each Pool and Operation. (C) Generating the sequence for $s_{Final} = 9$. Left: state assignment (blue). Right: sequence construction (magenta). Lists beside Pool A and mutagenize show which sequence or mutation each state selects. Inactive branches are shown in grey. (D) Style and name for the same sequence. Sequences are carried through the DAG as Seq objects, which hold the sequence string together with its .style. The style and prefix arguments are supplied to Operations; the resulting .style and name are held by Pools. Sequence characters are colored by their .style entries; where entries overlap, the last one applies. Panel D was generated with `_include_inline_styles=True`, which is omitted from the short code listing in panel A.

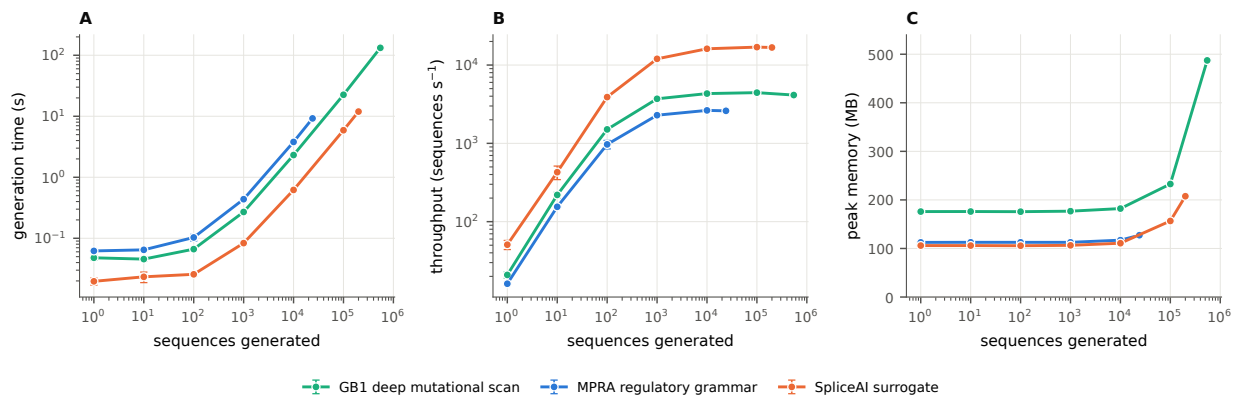


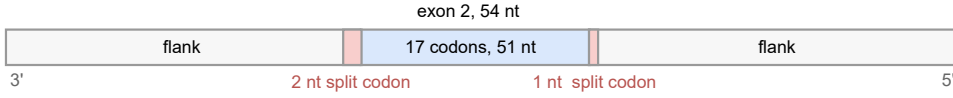
Figure S2: **Library generation scales linearly in time and memory.** (A) Wall-clock generation time, (B) throughput, and (C) peak resident memory against the number of sequences generated, for the three worked examples. Points are means of 5 replicate runs; error bars show ± 1 SD and are smaller than the plotting symbols at most points. Below $\sim 10^3$ sequences a fixed setup cost dominates, giving the plateau in (A) and the rise in (B); above 10^4 sequences throughput is constant, so generation time is linear in library size. Peak memory increases linearly, by 0.52–0.66 kB per sequence above a ~ 106 MB baseline.

Table S1: **Runtime and memory for the three worked examples.** Each example was generated at the size shown using a single core. Generation times are means $\pm$ SD over five replicate runs. Peak memory is maximum resident set size and includes the fixed memory used by Python and the example DAG. DAG construction does not depend on library size and is reported separately. The SpliceAI library comprises two matched 200,000-sequence pools (cryptic GT and disrupted GA) generated separately; values are for one pool, and the full library takes twice as long.

Example	Library size	Time (s)	Sequence/s	Memory (MB)	DAG (s)
GB1 deep mutational scan	547,230	132.24 ± 5.76	4,138	487	0.22
MPRA regulatory grammar	24,000	9.22 ± 0.22	2,602	127	0.06
SpliceAI surrogate	200,000	11.93 ± 0.27	16,758	208	0.05

A The library: BRCA1 exon 2, three mutation types, P5/P7 adaptors

chr17:43,115,634-43,115,878 minus strand, 245 nt



17 codons $\times$ (19 aa + 1 in-frame deletion + 1 stop) = 357 oligos

P5 20 nt + 245 nt + P7 21 nt = 286 nt (283 where a codon is deleted)

B PoolParty

```
import poolparty as pp

ADAPTOR_5 = "AATGATACGGCGACCACCGA"
ADAPTOR_3 = "TCGTATGCCGTCTTCTGCTTG"

# Extract region and annotate the target CDS
orf = pp.from_fasta(
    "chr17.fa", ("chr17", 43115633, 43115878, "-")
).annotate_orf(region_name="cds", extent=(100, 151), frame=1)

# Amino-acid substitutions
aa = orf.mutagenize_orf(
    region="cds", num_mutations=1,
    mutation_type="missense_only_first", mode="sequential")

# In-frame codon deletion scan
inframe = orf.deletion_scan_orf(
    deletion_codons=1, deletion_marker=None,
    region="cds", mode="sequential")

# Stop codon scan
stop = orf.insertion_scan_orf(
    "TGA", region="cds", replace=True, mode="sequential")

# Stack three pools, then add the adaptors
library = pp.join([ADAPTOR_5, pp.stack([aa, inframe, stop]),
    ADAPTOR_3])

library_df = library.generate_library()

Inputs: 1 FASTA
Outputs: oligos, with optional design card parameters that
produced them
```

C VaLiAnT

```
# Where to mutate and how
targeton.tsv
ref_chr      chr17
ref_strand    -
ref_start    43115634
ref_end      43115878
r2_start     43115726
r2_end       43115779
ext_vector    "25,25"
action_vector "(),(aa,inframe,stop),()"

# The transcript model, 51 records
ENST00000357654.9.gtf
...
CDS 43115726 43115779 . - 1
...

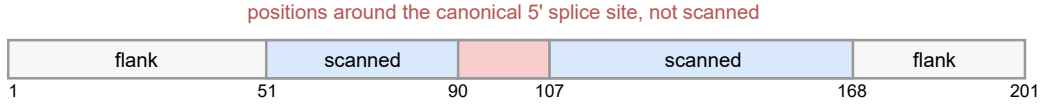
# The reference genome
chr17.fa 81 MB

# Adaptors, strand, and annotation as flags
$ valiant sge targeton.tsv ../ref/chr17.fa out \
  'homo sapiens' 'GRCh38' \
  --revcomp-minus-strand \
  --adaptor-5 ... --adaptor-3 ... \
  --gff ENST00000357654.9.gtf

Inputs: 3 files
Outputs: oligos, genomic coordinates,
consequence annotation, and a VCF
```

Figure S3: The same BRCA1 exon 2 variant library in PoolParty and VaLiAnT. (A) For each of the 17 complete codons in BRCA1 exon 2, the library contains substitutions to the other 19 amino acids, one stop-codon substitution, and one in-frame codon deletion. This gives 357 oligos. The split codons at the exon boundaries are left unchanged, and P5 and P7 adaptors are added to each sequence. (B) PoolParty implementation. The reference sequence is loaded from a FASTA file, and the user supplies the extent and reading frame of the 17 complete codons. Three ORF-aware Operations generate the amino-acid, stop-codon, and deletion scans. (C) VaLiAnT 4.0.0 implementation. VaLiAnT takes a coordinate and action table, a GTF annotation, and the reference chromosome as input. VaLiAnT gets the CDS phase from the GTF; in PoolParty, the user supplies the ORF extent and frame. The PoolParty code additionally calls `pp.set_genetic_code(valiant_codon_preference())` to match VaLiAnT's arginine codon choice; this setup line is omitted from panel B. When configured to use the same codon usage table, PoolParty and VaLiAnT gave exactly the same set of 357 sequences.

A The library: SMN2 exon 7, 2,000 cryptic 5'ss at 100 positions



2,000 cryptic 5' splice sites $\times$ 100 positions = 200,000 sequences

SCAN_POSITIONS = list(range(51,90)) + list(range(107,168))

B PoolParty

```
import poolparty as pp

# The 2,000 cryptic sites, as a pool
cryptic_pool = pp.from_seqs(
    cryptic_sites, mode="sequential")

# Replace one site at each position
library = pp.from_seq(target_region)\
    .replacement_scan(
        cryptic_pool,
        positions=SCAN_POSITIONS,
        mode="sequential")

# Final library, raw sequence strings
library_df = library.generate_library()
```

C tangermeme

```
import torch
from tangermeme.ersatz import substitute
from tangermeme.utils import one_hot_encode

# The 2,000 cryptic sites, as a tensor
cryptic_ohe = torch.stack(
    [one_hot_encode(site)
     for site in cryptic_sites])

# One copy of the region per cryptic site
target_ohe = (one_hot_encode(target_region)
               .unsqueeze(0)
               .expand(len(cryptic_sites), -1, -1))

# Substitute at each position,
# Final library, one-hot encoded tensor
library_ohe = torch.stack(
    [substitute(target_ohe, cryptic_ohe, start=p)
     for p in SCAN_POSITIONS])
```

Figure S4: The same cryptic splice-site perturbation library in PoolParty and tangermeme. (A) The library contains 2,000 cryptic 5'ss 9-mers, each substituted at 100 positions across a 201-bp region containing the canonical 5'ss of SMN2 exon 7, giving 200,000 sequences. Positions immediately around the canonical site are excluded. This is the GT arm of the library in main-text Fig. 4; the matched control arm is built the same way. (B) PoolParty implementation. The 2,000 sites form a Pool, and `replacement_scan` substitutes them at the 100 positions. The generated sequences are returned as strings in a table. (C) tangermeme 1.4.1 implementation. The target region and cryptic sites are one-hot encoded, substitutions are applied in one batch per position, and the final output is a tensor. After decoding this tensor, the two implementations gave the same set of 200,000 sequences.

Table S2: **Tools for designing DNA sequence libraries and for related sequence-design problems.** The final two rows group related tools; their members are named and cited individually.

Tool	Purpose / availability	Key features	Output
MPRA Design Tools [1]	Design MPRA libraries and estimate statistical power. R package (GitHub).	Barcode assignment with error-correcting sets; power analysis over effect size and sequencing depth	Barcoded constructs and statistical power estimates
MPRAnator [2]	Design MPRA libraries of motif arrangements and sequence variants. Web service.	Combinatorial placement of motifs; randomly sampled and combinatorial substitution sets; scrambled and reverse-complement controls	Barcoded library of sequences (FASTA)
Mutation Maker [3]	Design mutagenic oligos for building variant libraries in the laboratory. Web application (Docker); REST API.	Degenerate codons at user-listed sites; several substitutions combined in one reaction; de novo gene assembly with per-site variant frequencies	Primers, or gene fragments with the proportion of each variant to pool
Oligopool Calculator [4]	Add barcodes, primers, and spacers to a supplied variant set, and analyze the sequencing readout. Python package (PyPI); command line.	Constraint-aware barcodes, primers, and spacers; degenerate compression of a supplied variant set; off-target screening against a host genome	Synthesis-ready constructs; variant counts from sequencing reads
PoolParty	Design libraries that combine multiple variant types for laboratory assays or for probing genomic AI models. Python package (PyPI).	Libraries specified as a directed acyclic graph of Operations, allowing different mutagenesis schemes to be combined; sequences generated only on request	Library of sequences, each paired with a record of how it was constructed
tangermeme [5]	Generate perturbed sequence sets to probe cis-regulatory logic in deep learning models. Python package (PyPI).	Saturation mutagenesis of input sequences; insertion and removal of motifs across sets of sequences	Perturbed sequences, model predictions, or attribution scores
VaLiAnT [6]	Design and annotate libraries for saturation genome editing and cDNA deep mutational scanning. Command-line tool (source, Docker).	Saturation mutagenesis from genomic coordinates; several mutation types per target region; transcript-aware, including codons split across exons	Per-region libraries with metadata tables and VCF carrying variant consequence annotation
General-purpose toolkits (<i>Biopython</i> , <i>pydna</i> , <i>SeqPro</i>) [7–9]	Manipulate sequences, simulate cloning, prepare arrays for modeling. Python packages (PyPI).	Parsing, transformation, and file-format support	Transformed sequences
Sequence optimization tools (<i>CodonGenie</i> , <i>DNA Chisel</i> , <i>ledidi</i>) [10–12]	Optimize individual sequences for codon usage, synthesis constraints, or a model’s objective. Python packages (PyPI); web services.	Degenerate codons covering a target residue set; hard constraints and scored objectives combined in one problem; gradient-based editing guided by a predictive model	An optimized or edited sequence, or a degenerate codon

Table S3: **Capabilities of tools for designing DNA sequence libraries and for related sequence-design problems.** ● denotes a capability the tool provides; ◐ one provided with restrictions; ○ one the tool does not provide. Capabilities are counted only where the tool provides a dedicated operation, parameter, or mode.

	DNA Chisel	MPRA Design Tools	MPRAnator	Mutation Maker	Oligopool Calculator	PoolParty	tangermeme	ValiAnT
<i>Library Specification</i>								
Library object	○	◐	○	◐	◐	●	◐	○
Composable operations	◐	○	◐	○	●	●	◐	○
Mixed variant types in one library	○	●	●	◐	○	●	○	●
<i>Variant Generation</i>								
Saturation mutagenesis	○	○	○	○	◐	●	●	●
Randomly sampled variants	●	◐	●	○	○	●	○	○
Pairwise and higher-order variants	●	○	●	●	◐	●	◐	○
Model-guided variants	●	○	○	○	○	◐	●	○
<i>Variant Types</i>								
Codon / amino-acid substitutions	●	○	○	◐	○	●	○	●
Insertions and deletions	○	●	◐	○	○	●	●	●
Combinatorial multi-motif placement	○	○	●	○	○	●	●	○
Recombination	○	○	○	○	○	●	○	○
Shuffling	○	○	◐	○	○	●	●	○
<i>Physical Construction</i>								
Synthesis-constraint checking	●	◐	◐	○	●	◐	○	◐
Constraint-based optimization	●	◐	◐	●	●	○	○	○
Primer design	◐	○	○	●	●	○	○	○
<i>Metadata and Inspection</i>								
Per-sequence construction records	●	●	◐	●	●	●	○	●
Automatic naming	○	◐	●	●	◐	●	◐	●
Sequence styling	○	○	◐	○	○	●	○	○
<i>Genomic Integration</i>								
Genome coordinates	○	◐	●	○	○	◐	◐	●
Transcript / annotation aware	○	○	○	○	○	○	○	●

References

- [1] Ghazi AR, Chen ES, Henke DM, Madan N, Edelstein LC, Shaw CA. Design tools for MPRA experiments. *Bioinformatics*. 2018;34(15):2682–2683. <https://doi.org/10.1093/bioinformatics/bty150>.
- [2] Georgakopoulos-Soares I, Jain N, Gray JM, Hemberg M. MPRAinator: a web-based tool for the design of massively parallel reporter assay experiments. *Bioinformatics*. 2017;33(1):137–138. <https://doi.org/10.1093/bioinformatics/btw584>.
- [3] Hiraga K, Mejzlik P, Marcisin M, Vostrosablin N, Gromek A, Arnold J, et al. Mutation Maker, An Open Source Oligo Design Platform for Protein Engineering. *ACS Synth Biol*. 2021;10(2):357–370. <https://doi.org/10.1021/acssynbio.0c00542>.
- [4] Hossain A, Cetnar DP, LaFleur TL, McLellan JR, Salis HM. Automated Design of Oligopools and Rapid Analysis of Massively Parallel Barcoded Measurements. *ACS Synth Biol*. 2024;13(12):4218–4232. <https://doi.org/10.1021/acssynbio.4c00661>.
- [5] Schreiber J. tangermeme: A toolkit for understanding cis-regulatory logic using deep learning models. *bioRxiv*. 2025;p. 2025.08.08.669296. <https://doi.org/10.1101/2025.08.08.669296>.
- [6] Barbon L, Offord V, Radford EJ, Butler AP, Gerety SS, Adams DJ, et al. Variant Library Annotation Tool (VaLiAnT): an oligonucleotide library design and annotation tool for saturation genome editing and other deep mutational scanning experiments. *Bioinformatics*. 2022;38(4):892–899. <https://doi.org/10.1093/bioinformatics/btab776>.
- [7] Cock PJA, Antao T, Chang JT, Chapman BA, Cox CJ, Dalke A, et al. Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics*. 2009;25(11):1422–1423. <https://doi.org/10.1093/bioinformatics/btp163>.
- [8] Pereira F, Azevedo F, Carvalho Â, Ribeiro GF, Budde MW, Johansson B. Pydna: a simulation and documentation tool for DNA assembly strategies using python. *BMC Bioinformatics*. 2015;16(1):142. <https://doi.org/10.1186/s12859-015-0544-x>.
- [9] Klie A, Laub D. SeqPro (Sequence processing toolkit); 2023. Software. Available from: <https://github.com/ML4GLand/SeqPro>.
- [10] Swainston N, Currin A, Green L, Breitling R, Day PJ, Kell DB. CodonGenie: optimised ambiguous codon design tools. *PeerJ Comput Sci*. 2017;3:e120. <https://doi.org/10.7717/peerj-cs.120>.
- [11] Zulkower V, Rosser S. DNA Chisel, a versatile sequence optimizer. *Bioinformatics*. 2020;36(16):4508–4509. <https://doi.org/10.1093/bioinformatics/btaa558>.
- [12] Schreiber J, Lorbeer FK, Heinzl M, Lu YY, Stark A, Noble WS. Programmatic design and editing of cis-regulatory elements. *bioRxiv*. 2025;p. 2025.04.22.650035. <https://doi.org/10.1101/2025.04.22.650035>.